

Networking 101

If you never knew the difference between a router and a modem then this will give you a little heads up on the what's what in the world of networking.

We've all been on the internet, but have you ever wondered as to what constitutes this mammoth structure? We'll let's start from a small scale - two computers connected to each other. There are two common ways of doing this. One is to use an ethernet patch cord and connect it to the other computer's ethernet port. The two machines have now formed a small network. Such a network is useful for impromptu file transfers and serves a single purpose only. The other way is to connect both computers to a router/hub/switch and create a LAN. Now not only can you transfer files over to each other but the router/hub/switch can be connected to another network or to the internet.

There are different types of networks, each defined over the area and purpose that it serves. There is no fixed distance parameter over which each network is defined but a hierarchy in the increasing order of size is given below.

PAN - Personal Area Network < LAN - Local Area Network < MAN - Metropolitan Area Network < WAN - Wide Area Network

Packets

Data is sent across networks in an encapsulated form. Each unit of data is called a packet and each packet is of a fixed size depending upon the protocol used. The entire network is made of connecting devices and wires. The types of devices include routers, modems,

switches, hubs, bridges, repeaters and the wires include Cat5, Cat5e, Cat6, Fibre, Coaxial. Each device and cable is suited for a few specific purposes. Millions of networking devices are connected to each other using these devices (and others) to form the internet.

Addresses

Each device on a network has an address assigned to it. Right now we use IPv4 addresses which are of the form xxx.xxx.xxx.xxx where xxx ranges from 000 to 255. IPv4 allows for 2^{32} (4,294,967,296) addresses. This may seem very miniscule compared to the number of networking enabled devices on the internet, but each of these addresses can have an entire network behind them. IPv4 addresses have been exhausted and IPv6 addresses are now being assigned.

Devices

Router

A router forwards packets that it receives to their destination. The router examines each packet to get the destination address and forwards the packets to it. Take the router we have in our homes, multiple computers can be connected to a router but it makes sure that your traffic only comes to you and not to anyone else on the network.

Modem

Stands for modulation-demodulation. It's a simple device that encodes and decodes data onto a carrier signal. A modulated signal is capable of losing less data due to attenuation over large distances.

Hub

A device that connects a few routers



together, thus, essentially connecting a few networks together to create a small network.

Bridge

If two networks deal in different forms of packets or protocol then a bridge connects the two networks together. It checks the packet's destination to figure out if it needs to be sent across the bridge or not.

Switch

Differs slightly from a bridge in the sense that it is much more efficient in operation.

Repeater

Over large distances a communication signal tends to lose its strength and a lot of unwanted noise creeps in. A repeater takes this signal and regenerates it. Different types of signals have different distances at which a repeater is required. Eg. Cat5e and Cat6 require repeaters at a distance of 100 meters.

Standards

There are different protocols for connectivity and each is called a standard. The normal ethernet is standard IEEE 802.3, WiFi is IEEE 802.11 and IEEE 802.15.1 is Bluetooth. All protocols adhere to two standard models - OSI (Open Systems Interconnect) and TCP/IP model.

Client-Server Model

A majority of networking protocols are implemented in this manner. One node (server) manages a host of other nodes (clients). These servers in turn become clients for other servers which are at a higher place in the hierarchy. 

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